

How Age, Driving Experience, and Gender Interact in Predicting Dangerous Driving Behavior

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Abstract

While increasing age has been consistently associated with a safer driving style, the role of driving experience and its interplay with age remains insufficiently understood. Using a large community-based sample of drivers, this study aimed to clarify the relative, joint, and interactive contributions of age, driving experience, and gender to the propensity to engage in dangerous driving behaviors. The frequency of such behaviors was assessed using the Dula Dangerous Driving Index (DDDI). The sample included 1,882 drivers (50.9% men) with a mean age of 38.11 years (range 18–79), a mean driving experience of 15.94 years, and a mean annual mileage of 6,000–9,000 km. Results indicated that increasing age was associated with a reduced propensity for dangerous driving behavior, but only after sufficient driving experience had been acquired. Furthermore, greater driving experience predicted higher dangerous driving behavior in younger and middle-aged drivers, while remaining unrelated in older drivers. Notably, this pattern was more pronounced among men than women. The observed interaction may be explained by the fact that novice drivers, regardless of age, possess lower driving skills due to their limited experience, which may encourage a more cautious driving style, offsetting the greater risk-taking typically observed in youth. However, once sufficient experience is gained, younger drivers demonstrate a marked tendency toward greater risk-taking behavior. These findings emphasize the need for multifactorial approaches that account for the interactions among contributing variables to guide the implementation of effective road-safety measures.

Keywords: Dangerous driving; Risk-taking behavior; Age; Driving experience; Gender differences; DDDI.

1. Introduction

Traffic crashes remain a major public health concern worldwide, with dangerous driving practices constituting a leading cause of crashes and fatalities (World Health Organization, 2023). Young drivers, particularly young men, account for a high percentage of individuals involved in such risk behaviors and crashes (Cassarino & Murphy, 2018; de Winter & Dodou, 2010; Fergusson et al., 2003; McCartt, 2009). This overrepresentation is often attributed to a combination of the higher risk propensity characteristic of youth (Hatfield & Fernandes, 2009; Jonah, 1986; Voogt et al., 2014), together with limited driving skills resulting from insufficient experience, as obtaining a driver's license usually coincides with this stage of life (Bjørnskau & Sagberg, 2005; Deery, 1999; Lee et al., 2008; McCartt et al., 2009; McKenna et al., 2006). A detailed understanding of how these factors influence driving behavior is crucial for designing effective prevention strategies and enhancing road safety. In this regard, although age and driving experience have previously been studied together as predictors of dangerous driving (Brown & Groeger, 1988; de Winter & Dodou, 2010; Huang & Winston, 2011; Maycock et al., 1991; McCartt et al., 2009), the nature of their interrelation remains unclear.

The literature has consistently reported that the propensity to take risks behind the wheel decreases with age, a pattern that is particularly evident when comparing drivers aged 18–24 with older drivers (de Winter & Dodou, 2010; Hatfield & Fernandes, 2009; Jonah, 1990; Rhodes & Pivik, 2011; Sánchez-López et al., 2024; Voogt et al., 2014; Yousaf et al., 2024). The heightened tendency to engage in dangerous driving behavior during youth has been associated with elevated levels of impulsivity, greater risk propensity and sensation seeking, perceived invulnerability, optimism bias, difficulties in emotion regulation, and susceptibility to peer influence (Aluja et al., 2023; DeJoy, 1992; Gheorghiu et al., 2015; Hatfield et al. 2017; Huang & Winston, 2011; Ramos-Moreno et al., 2025; Pizzo et al., 2024; Rhodes et al., 2015; Starkey & Isler, 2016).

Neurodevelopmental theories further support this perspective, proposing that adolescence and early adulthood are characterized by increased novelty- and sensation-seeking behaviors, without a parallel maturation of the self-regulatory capacities needed to manage these impulses effectively (Steinberg, 2008; 2010). This developmental imbalance

may help explain the peak in risky, and sometimes antisocial, driving behaviors observed among young drivers. Empirical data on traffic crash rates support this relationship, showing a marked decrease in crash frequency among drivers aged 35+ compared with those aged 20–24. Furthermore, although the most substantial decline occurs in the early stages of adulthood, the downward trend in crash rates continues across older age groups (McCartt et al., 2009; National Center for Statistics and Analysis, 2025). Taken together, these findings highlight the critical influence of age-related factors on driving safety.

Driving experience, in turn, contributes to the development of critical skills such as vehicle control, visual scanning, and hazard and risk perception, which improve progressively through real-world exposure and training (Borowsky et al., 2010; Deery, 1999; Ge et al., 2017; McKnight & McKnight, 2003; Megías et al., 2018; Lehtonen et al. 2014). Regardless of the driver's age, inexperience is associated with an increased risk of crashes due to the insufficient development of these critical driving skills. Evidence indicates that both young and older novice drivers initially exhibit high crash rates, which decline markedly with practice, a reduction unlikely to be explained solely by maturity or other personal factors (Huang & Winston, 2011; Langford et al., 2006; Maycock et al., 1991; McCartt et al., 2009). Some studies even suggest that driving experience may be a more decisive factor than age itself in preventing road crashes (McCartt et al., 2009; McKnight & McKnight, 2003).

As outlined above, the link between driving experience and crash probability appears to be well established; however, the relationship between driving experience and engagement in dangerous driving behaviors — the primary focus of the present research — remains less clear, with studies yielding mixed findings. On the one hand, the literature indicates that inexperienced drivers are more susceptible to driving risks than their more experienced counterparts, due to a limited ability to effectively detect and anticipate potential hazards. As these drivers are still in a learning phase, developing driving skills and reducing errors are ongoing processes (Deery, 1999; McKenna et al., 2006; Lee et al., 2008). Consistent with this view, Waller et al. (2001), in a study involving 13,809 novice drivers in Michigan, found that the incidence of serious traffic violations declined with increasing driving experience, with an approximate 8% reduction for each year of licensure during the initial years of driving. On the other hand, as drivers become more experienced, their greater self-efficacy

may lead to riskier attitudes and a higher incidence of traffic violations due to increased confidence in their driving skills. Thus, although experience can reduce certain risks by enhancing skills, it may simultaneously promote more risk-taking behavior (Catchpole et al., 2005; de Winter & Dodou, 2010; Jing et al., 2023; Machado-León et al., 2016; Qu et al., 2014). Supporting this notion, de Winter and Dodou's (2010) meta-analysis reported that driving violations are positively associated with driving experience. Evidence from several longitudinal studies has also shown that, during the early stages of a driving career, driving style tends to become riskier with increasing experience, measured as time since obtaining a driving license, regardless of age or gender (Rowe et al., 2013; Wells et al., 2008).

Gender constitutes another important factor to consider in this context. Numerous studies indicate that, compared with women, men are generally more prone to taking risks on the road, engaging in aggressive and impulsive behaviors, committing traffic violations, and incurring higher numbers of fines and crashes. In contrast, women tend to adopt more responsible driving attitudes, which act as protective factors against dangerous driving behaviors (Cordellieri et al., 2016; de Winter & Dodou, 2010; González-Iglesias et al., 2012; Megías-Robles et al., 2022; Navas et al., 2019; Oltedal & Rundmo, 2006).

Despite extensive research on the roles of age, experience, and gender, considerable debate persists in the literature over the relative importance of these factors in promoting dangerous driving behaviors. Some authors argue that developmental and psychological characteristics associated with age are the most influential factors (Harbeck et al., 2017; Laapotti et al., 2001; Ventsislavova et al., 2021), whereas others contend that the accumulation of driving experience represents the primary determinant, regardless of a driver's chronological age (Ge et al., 2017; Huang & Winston, 2011; Rowe et al., 2013; Waller et al., 2001). However, we believe a crucial aspect of this debate that is generally overlooked is the potential interaction effect of these variables. The lack of driving skills among novice drivers is common across all ages and may moderate age-related effects. Lower proficiency, together with the resulting reduced sense of self-efficacy, may lead to more cautious driving, potentially counteracting the higher risk-taking behavior that typically characterizes younger drivers.

Building on the previous discussion, the present study aimed to take a step further in understanding how age and driving experience are associated with dangerous driving behavior. Rather than focusing solely on their individual associations or treating them as covariates to be controlled for, we sought to examine the possibility of an interaction between these variables. This analysis was complemented by examining gender-related differences and their potential moderating effects on the relationship between age, experience, and dangerous driving. We hypothesized the following: (H1) In terms of individual variables, younger drivers will exhibit more dangerous driving behaviors than older drivers; with respect to driving experience, however, no clear directional hypothesis can be formulated regarding its relationship with dangerous driving behavior, given the mixed findings reported in the literature. Nevertheless, we propose that (H2) the level of driving experience may influence the relationship between age and dangerous behavior, as discussed above; and (H3) men will exhibit more dangerous driving behaviors than women. The interaction effects involving gender, age, and driving experience will be explored.

To test these hypotheses, we assessed drivers' propensity for dangerous behavior using the Dula Dangerous Driving Index (DDDI; risky driving and aggressive driving subscales). Levels of driving experience were recorded based on the number of years holding a driving license and annual car mileage. Analyses were conducted on two different samples: the first comprised a broad community sample, while the second was restricted to a subsample of drivers who had held a driving license for 10 years or less. This reduced subsample was selected because the effects of experience are expected to be most pronounced during the early stages of a driving career. Once sufficient experience is gained, changes in driving behavior related to skill acquisition are less likely to occur. Moreover, this restriction ensures a more balanced distribution of participants in terms of experience, preventing highly experienced drivers from being concentrated exclusively among older individuals. Thus, by focusing on drivers in the initial years of driving, the influence of experience and its potential interaction with age may be examined more precisely.

2. Method

2.1. Participants

The sample comprised 1,882 drivers. All participants held a valid car driving license and were recruited within the context of larger studies. The mean age was 38.11 years ($SD = 14.37$; range = 18–79), with 957 men (50.9%). Participants had held their driving license for an average of 15.94 years ($SD = 13.48$; range = 0–61) and reported a mean annual mileage score of 5.45 ($SD = 2.80$), corresponding to an estimated driving distance of approximately 6,000–9,000 km per year according to the scale used in the study (see Procedure section).

Participants were recruited through advertisements at the Universities of Málaga and Granada, with undergraduate students from these institutions assisting in the recruitment process to ensure a balanced sample across age and gender. The study was conducted in accordance with the ethical principles of the Declaration of Helsinki (World Medical Association, 2013). Prior to participation, individuals were informed about the study procedures, the voluntary nature of their involvement, and the anonymity of their data. Subsequently, they provided electronic informed consent. Ethical approval for this project was granted by the Research Ethics Committee of the University of Málaga (approval number: 14–2019-H).

2.2. Procedure and Instruments

Before completing the DDDI, participants provided sociodemographic information, including gender, age, the number of years they had held a driving license (hereafter referred to as years holding a driving license), and annual car mileage. Following the procedure used by Megías-Robles et al. (2022) and Megías et al. (2018), annual car mileage was assessed using a 12-category scale: 1 = “Nothing,” 2 = “Less than 1,000 km,” 3 = “1,000–3,000 km,” 4 = “3,000–6,000 km,” 5 = “6,000–9,000 km,” 6 = “9,000–12,000 km,” 7 = “12,000–15,000 km,” 8 = “15,000–18,000 km,” 9 = “18,000–21,000 km,” 10 = “21,000–30,000 km,” 11 = “30,000–40,000 km,” and 12 = “More than 40,000 km.”

The Dula Dangerous Driving Index (DDDI) is a self-report instrument employed to measure dangerous driving behavior (Dula & Ballard, 2003; Sánchez-López et al., 2024; Willemsen et al., 2008). It comprises 28 items divided into three subscales: Risky Driving

(DDDI-RD; 12 items), Aggressive Driving (DDDI-AD; 7 items), and Negative Cognitive/Emotional Driving (DDDI-NCED, formerly referred to as NED in some prior studies; 9 items). Participants are asked to indicate the frequency with which they engage in specific driving behaviors, responding to each item on a 5-point Likert scale (1 = 'Never', 2 = 'Rarely', 3 = 'Sometimes', 4 = 'Often', 5 = 'Always'). For the computation of mean scores for each subscale, values were rescaled to a 0–4 range to facilitate interpretation. The Spanish version of the DDDI has demonstrated good internal consistency (Cronbach's $\alpha = .86$ for DDDI-RD, $.82$ for DDDI-AD, and $.82$ for DDDI-NCED; Sánchez-López et al., 2024). In the present sample, internal consistency was acceptable, with Cronbach's $\alpha = .82$ for DDDI-RD, $.78$ for DDDI-AD, and $.79$ for DDDI-NCED.

2.3. Data analysis

The analyses focused on the DDDI-RD and DDDI-AD subscales. The DDDI-NCED subscale was not considered, as it do not directly target intentional dangerous driving behaviors. However, for readers who may be interested, the main findings for this subscale are briefly summarized at the end of the Results section, and the full results are reported in the supplementary material (see Section 1).

Prior to conducting analyses, the assumptions of normality and homoscedasticity of the residuals were evaluated for the multiple regression models to ensure their robustness. Visual inspection of Q–Q plots of the standardized residuals for the DDDI subscales revealed deviations from normality. This observation was supported by significant Kolmogorov-Smirnov and Shapiro-Wilk tests ($p < .001$). Skewness values ranged from 0.82 to 1.15 and excess kurtosis values ranged from 0.73 to 1.53. Regarding homoscedasticity, inspection of scatterplots of standardized residuals against predicted values showed no evidence of heteroscedasticity. Given the violation of the normality assumption, subsequent analyses were conducted using a bias-corrected bootstrapping procedure (5,000 resamples) to obtain robust parameter estimates. Regression coefficients were interpreted based on 95% bootstrap confidence intervals, with intervals not including zero indicating a significant relationship.

As explained in the Introduction, the data analysis was conducted in two phases. The first phase included the full sample of participants, whereas the second phase focused on a

reduced subsample comprising only drivers who had held a driving license for 10 years or less.

Descriptive statistics and bootstrapped Pearson correlations were computed for the study variables in both phases. Next, in accordance with our hypotheses, a series of multiple linear regression analyses were conducted, with each DDDI subscale serving as the criterion variable. In the first phase (full sample), the predictors were Gender, Age, Years holding a driving license, Annual car mileage, the two-way interaction terms Age $\times$ Years holding a driving license and Age $\times$ Annual car mileage, as well as all corresponding interaction terms involving gender (Gender $\times$ Age, Gender $\times$ Years holding a driving license, Gender $\times$ Annual car mileage, Gender $\times$ Age $\times$ Years holding a driving license, and Gender $\times$ Age $\times$ Annual car mileage). In the second phase (reduced subsample: participants who had held a driving license for 10 years or less), the predictors included were Gender, Age, Experience, Age $\times$ Experience, and the interaction terms involving Gender (Gender $\times$ Age, Gender $\times$ Experience, and Gender $\times$ Age $\times$ Experience). Note that in this phase, the variables Years holding a driving license and Annual car mileage were replaced by a composite Experience variable to assess driving experience. This variable was calculated by multiplying the number of years holding a driving license by annual car mileage. We consider that this measure may provide an assessment of driving experience that is closer to total lifetime mileage. In the first phase, potential fluctuations in annual mileage over time — particularly among individuals with many years of driving experience — discouraged us from using this composite measure. For readers who may be interested, the results of the second phase using Years holding a driving license and Annual car mileage as separate predictors are provided in the supplementary material (see Section 2).

When significant interactions were identified in the multiple regression analyses, follow-up analyses using simple slopes were conducted to examine the pattern of these interactions. Specifically, the relationship between a given predictor and the criterion variable was examined at the 16th (low), 50th (medium), and 84th (high) percentiles of the other predictor.

All analyses were conducted using SPSS version 22 (IBM Corporation, Armonk, NY, USA) and the PROCESS macro version 4.2 (Hayes, 2022).

3. Results

3.1. Full Sample

Descriptive statistics for the full sample of drivers are presented in Table 1. Bootstrapped Pearson correlations (5,000 resamples) among the DDDI scores and drivers' sociodemographic characteristics are presented in Table 2.

Table 1. Means and standard deviations of the study variables in the full sample, for all participants and by gender.

	Mean (<i>SD</i>) Total sample	Mean (<i>SD</i>) Men	Mean (<i>SD</i>) Women
Age	38.11 (14.37)	37.86 (14.85)	38.38 (13.87)
Years driving license	15.94 (13.48)	16.60 (14.26)	15.26 (12.59)
Annual car mileage	5.45 (2.80)	6.21 (2.84)	4.67 (2.53)
DDDI-RD	0.65 (0.48)	0.77 (0.53)	0.52 (0.39)
DDDI-AD	0.67 (0.57)	0.78 (0.61)	0.56 (0.50)

Abbreviations: DDDI RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI AD = Dula Dangerous Driving Index Aggressive Driving subscale.

Table 2. Bootstrapped Pearson correlation coefficients among the study variables in the full sample, with *r* values and corresponding 95% confidence intervals reported (in square brackets).

	1	2	3	4
1. Age	—			
2. Years driving license	.90* [.88, .91]	—		
3. Annual car mileage	.13* [.09, .18]	.18* [.14, .22]	—	
4. DDDI-RD	-.22* [-.26, -.17]	-.17* [-.21, -.12]	.15* [.11, .19]	—
5. DDDI-AD	-.07* [-.11, -.03]	-.04 [-.08, .01]	.15* [.11, .19]	.60* [.56, .63]

* Confidence interval does not include zero.

Abbreviations: DDDI-RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI-AD = Dula Dangerous Driving Index Aggressive Driving subscale.

The multiple linear regression analysis for DDDI-RD in the full sample indicated that the model accounted for 14% of the variance in risky driving behavior ($R = .38$; see Table 3). The predictors that reached significance were Gender ($B = -0.208$, $SE = 0.034$, 95% $CI [-.273, -.144]$), Age ($B = -0.008$, $SE = 0.003$, 95% $CI [-.013, -.003]$), Annual car mileage ($B = 0.018$, $SE = 0.006$, 95% $CI [.005, .030]$), and the interaction terms Age $\times$ Annual car mileage ($B = -0.002$, $SE = 0.001$, 95% $CI [-.003, -.001]$) and Age $\times$ Annual car mileage $\times$ Gender ($B = 0.001$, $SE = 0.001$, 95% $CI [.0003, .0025]$). Follow-up analyses of the second-order interaction revealed that, among men, Age was negatively related to DDDI-RD scores at medium ($B = -0.007$, $SE = 0.003$, 95% $CI [-0.013, -0.002]$) and high levels of annual car mileage ($B = -0.015$, $SE = 0.003$, 95% $CI [-0.020, -0.008]$), but not at low levels (95% CI included zero). Annual car mileage was positively related to DDDI-RD scores when Age was low ($B = 0.050$, $SE = 0.008$, 95% $CI [0.034, 0.066]$) and medium ($B = 0.020$, $SE = 0.005$, 95% $CI [0.009, 0.031]$), whereas this relationship was not significant at high values of Age (95% CI included zero). For women, Age was negatively related to DDDI-RD regardless of Annual car mileage (low: $B = -0.007$, $SE = 0.002$, 95% $CI [-0.012, -0.002]$; medium: $B = -0.08$, $SE = 0.002$, 95% $CI [-0.013, -0.003]$; high: $B = -0.009$, $SE = 0.003$, 95% $CI [-0.016, -0.003]$) and Annual car mileage was positively related to DDDI-RD regardless of age (low: $B = 0.035$, $SE = 0.009$, 95% $CI [0.016, 0.053]$; medium: $B = 0.027$, $SE = 0.006$, 95% $CI [0.015, 0.039]$; high: $B = 0.019$, $SE = 0.009$, 95% $CI [0.014, 0.036]$). In addition, women reported significantly lower levels of risky driving behavior than men across all conditions (all 95% CI s excluded zero). See Figure 1A for a visual representation of the Age $\times$ Annual car mileage $\times$ Gender interaction.

Regarding DDDI-AD scores, the multiple linear regression analysis revealed that the model accounted for 6% of the variance in aggressive driving behavior ($R = 0.25$; see Table 3). The predictors that reached significance were Gender ($B = -0.185$, $SE = 0.045$, 95% $CI [-0.269, -0.094]$), Annual car mileage ($B = 0.024$, $SE = 0.007$, 95% $CI [0.009, 0.037]$), and the Age $\times$ Annual car mileage interaction ($B = -0.002$, $SE = 0.000$, 95% $CI [-0.003, -0.001]$). Follow-up analyses of this interaction indicated that Age was negatively related to DDDI-AD scores at high levels of annual car mileage ($B = -0.010$, $SE = 0.004$, 95% $CI [-0.017, -0.003]$), but not at medium or low levels (95% CI included zero). Annual car mileage was positively related to DDDI-AD scores when Age was low ($B = 0.052$, $SE = 0.010$, 95% CI

[0.033, 0.072]) and medium ($B = 0.025$, $SE = 0.007$, 95% CI [0.012, 0.039]), but not when Age was high (95% CI included zero). Regarding Gender, women reported lower levels of aggressive driving than men across all conditions (95% CI s excluded zero). Figure 1B depicts a visual representation of the interaction.

Table 3. Results for the multiple linear regression analyses predicting DDDI-RD and DDDI-AD in the full sample.

Regression model	Predictors	R^2	B	SE	Confidence interval 95%	
					Lower limit	Upper limit
Criterion: DDDI-RD		.14				
	Gender		-0.208*	0.034	-.273	-.144
	Age		-0.008*	0.003	-.013	-.003
	Years driving license		-0.000	0.003	-.006	.006
	Annual car mileage		0.018*	0.006	.005	.030
	Gender x Age		0.000	0.003	-.007	.006
	Gender x Years driving license		0.001	0.004	-.006	.009
	Gender x Annual car mileage		0.009	0.008	-.007	.025
	Age x Years driving license		0.000	0.000	-.000	.000
	Age x Annual car mileage		-0.002*	0.000	-.003	-.001
	Gender x Age x Years driving license		-0.000	0.000	-.000	.000
	Gender x Age x Annual car mileage		0.001*	0.001	.000	.002
Criterion: DDDI AD		.06				
	Gender		-0.185*	0.045	-.269	-.094
	Age		-0.004	0.004	-.011	.003
	Years driving license		0.003	0.004	-.005	.011
	Annual car mileage		0.024*	0.007	.009	.037
	Gender x Age		-0.004	0.005	-.014	.005
	Gender x Years driving license		0.002	0.005	-.008	.013
	Gender x Annual car mileage		-0.003	0.009	-.022	.016
	Age x Years driving license		-0.000	0.000	-.000	.000
	Age x Annual car mileage		-0.002*	0.000	-.003	-.001
	Gender x Age x Years driving license		-0.000	0.000	-.000	.000
	Gender x Age x Annual car mileage		0.001	0.001	-.000	.002

* Confidence interval does not include zero.

Abbreviations: DDDI-RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI-AD = Dula Dangerous Driving Index Aggressive Driving subscale.

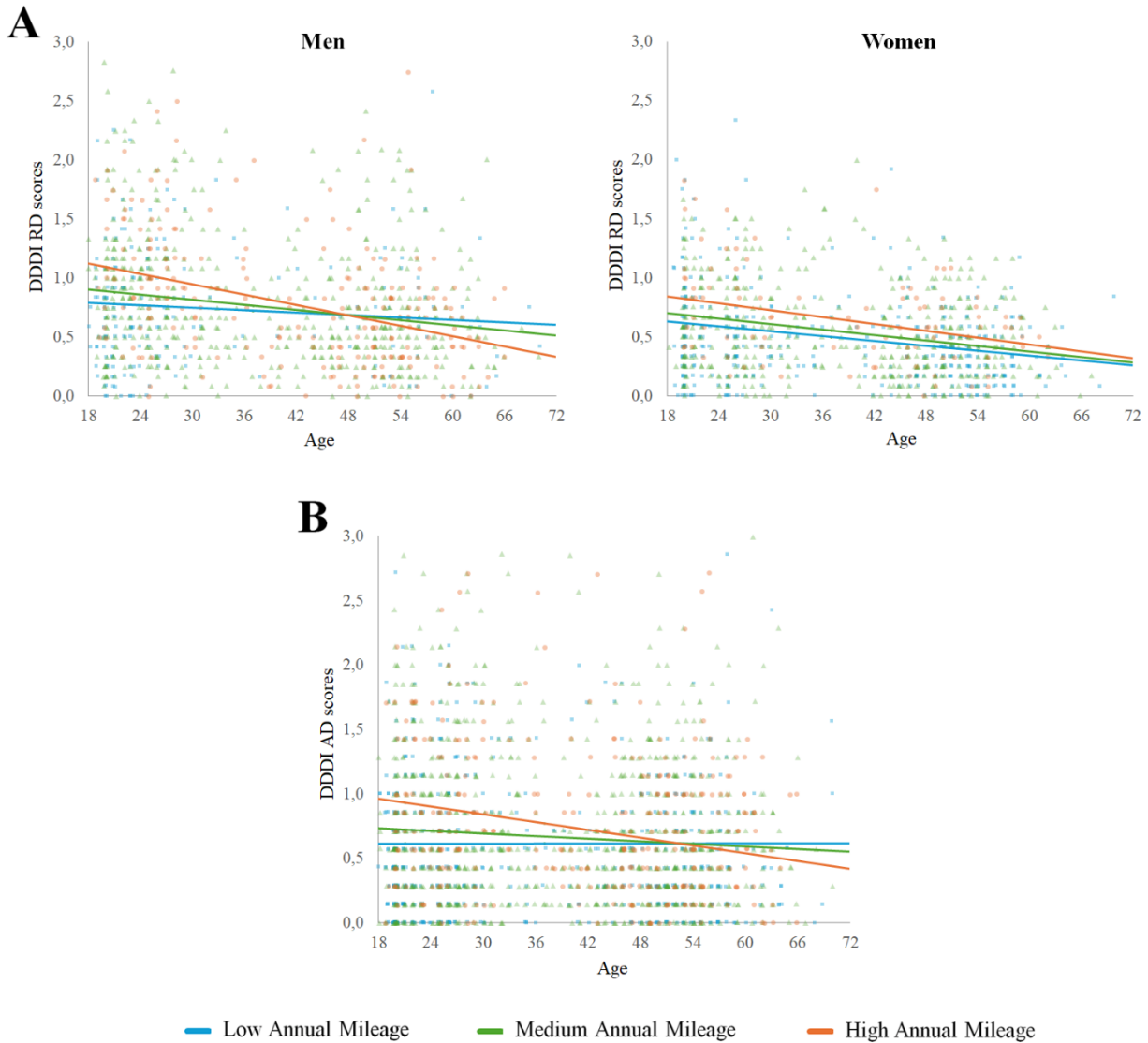


Figure 1. Scatter plots illustrating the significant interactions from the multiple regression models predicting DDDI scores, with points representing observed values and regression lines representing predicted values. Panel A shows the Gender $\times$ Age $\times$ Annual car mileage interaction for the DDDI-RD subscales, displayed separately for men (left) and women (right). Panel B shows the Age $\times$ Annual car mileage interaction for the DDDI-AD subscale. In both panels, the Y-axis represents DDDI scores and the X-axis represents age. Observed data points are coloured according to annual car mileage groups based on percentiles: low (≤ 16 th percentile, blue), medium (17th–83rd percentile, green), and high (≥ 84 th percentile, orange). Regression lines, based on predicted values from each model, illustrate the simple slopes for each mileage group calculated from percentiles (16th, 50th, and 84th) and are plotted in the same colour as the observed data points corresponding to each group.

3.2. Reduced subsample: Drivers Holding a License for 10 years or Less

Descriptive statistics for the reduced subsample of drivers with 10 years or less holding a driving license are presented in Table 4. Bootstrapped Pearson correlations (5,000 resamples) among the DDDI subscale scores and drivers' sociodemographic characteristics are presented in Table 5.

Table 4. Means and standard deviations of the study variables in the reduced subsample (≤ 10 years of driving license), for all participants and by gender.

	Mean (<i>SD</i>) Total sample	Mean (<i>SD</i>) Men	Mean (<i>SD</i>) Women
Age	25.93 (7.99)	25.11 (6.85)	26.79 (8.96)
Years driving license	3.76 (3.15)	3.77 (3.06)	3.74 (3.25)
Annual car mileage	4.98 (2.67)	5.38 (2.76)	4.55 (2.52)
Experience	20.29 (22.38)	22.17 (23.56)	18.30 (20.92)
DDDI-RD	0.74 (0.52)	0.89 (0.56)	0.59 (0.43)
DDDI-AD	0.69 (0.60)	0.79 (0.63)	0.58 (0.53)

Abbreviations: DDDI RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI AD = Dula Dangerous Driving Index Aggressive Driving subscale.

Table 5. Bootstrapped Pearson correlation coefficients among the study variables in the reduced subsample (≤ 10 years of holding a driving license), with *r* values and corresponding 95% confidence intervals reported (in square brackets).

	1	2	3	4	5
1. Age	—				
2. Years driving license	.37* [.31, .44]	—			
3. Annual car mileage	.05 [-.02, .11]	.19* [.12, .25]	—		
4. Experience	.27* [.21, .34]	.79* [.76, .81]	.63* [.58, .67]	—	
5. DDDI-RD	-.14* [-.20, -.09]	.03 [-.03, .10]	.25* [.18, .31]	.13* [.06, .19]	—
6. DDDI-AD	-.04 [-.10, .01]	.06* [.00, .13]	.19* [.13, .26]	.11* [.05, .18]	.58 [.53, .63]

* Confidence interval does not include zero.

Abbreviations: DDDI-RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI-AD = Dula Dangerous Driving Index Aggressive Driving subscale.

For drivers holding a driving license for 10 years or less, the multiple linear regression model for DDDI-RD accounted for 12% of the variance (see Table 6). The predictors identified as significant were Gender ($B = -0.285$, $SE = 0.034$, 95% $CI [-0.352, -0.218]$), Age ($B = -0.012$, $SE = 0.003$, 95% $CI [-0.019, -0.006]$), and Experience ($B = 0.005$, $SE = 0.001$, 95% $CI [0.002, 0.007]$), as well as the interaction terms of Age $\times$ Experience ($B = -0.0004$, $SE = 0.0002$, 95% $CI [-0.0008, -0.0002]$), and Age $\times$ Experience $\times$ Gender ($B = 0.0004$, $SE = 0.0002$, 95% $CI [0.0001, 0.0008]$). Follow-up analyses of the second-order interaction revealed that, among men, Age was negatively related to DDDI-RD scores at medium ($B = -0.009$, $SE = 0.004$, 95% $CI [-0.016, -0.001]$) and high levels of experience ($B = -0.022$, $SE = 0.005$, 95% $CI [-0.033, -0.012]$), but not at low levels (95% CI included zero). Experience was positively related to DDDI-RD scores in the three Age groups (low: $B = 0.007$, $SE = 0.002$, 95% $CI [0.004, 0.010]$; medium: $B = 0.005$, $SE = 0.001$, 95% $CI [0.003, 0.008]$; high: $B = 0.003$, $SE = 0.001$, 95% $CI [0.001, 0.005]$). For women, Age showed a negative relationship with DDDI-RD regardless of the level of Experience (low: $B = -0.009$, $SE = 0.003$, 95% $CI [-0.015, -0.003]$; medium: $B = -0.009$, $SE = 0.003$, 95% $CI [-0.015, -0.004]$; high: $B = -0.010$, $SE = 0.004$, 95% $CI [-0.018, -0.003]$), whereas Experience showed a positive relationship with DDDI-RD regardless of Age (low: $B = 0.003$, $SE = 0.002$, 95% $CI [0.0001, 0.0068]$; medium: $B = 0.003$, $SE = 0.001$, 95% $CI [0.0005, 0.0061]$; high: $B = 0.003$, $SE = 0.001$, 95% $CI [0.0007, 0.0055]$). Across all conditions, women reported significantly lower levels of risky driving behavior than men (all 95% CI s did not include zero).

Regarding DDDI-AD scores, the multiple linear regression analysis using the sample of drivers holding a license for 10 years or less revealed that the model accounted for 5% of the variance (see Table 6). Significant predictors were Gender ($B = -0.201$, $SE = 0.041$, 95% $CI [-0.282, -0.118]$), Experience ($B = 0.005$, $SE = 0.001$, 95% $CI [0.002, 0.007]$), and the Age $\times$ Experience interaction ($B = -0.0005$, $SE = 0.0002$, 95% $CI [-0.0008, -0.0001]$). Follow-up analyses of this interaction showed that Age was negatively related to aggressive driving behavior when Experience was high ($B = -0.015$, $SE = 0.006$, 95% $CI [-0.028, -0.003]$), but not when Experience was low or medium (95% CI s included zero). Experience was positively related to aggressive driving behavior in the three Age groups (low: $B = 0.007$, $SE = 0.002$, 95% $CI [0.004, 0.011]$; medium: $B = 0.005$, $SE = 0.001$, 95% $CI [0.003, 0.008]$; high: $B = 0.003$, $SE = 0.001$, 95% $CI [0.0006, 0.0054]$).

Table 6. Results for the multiple linear regression analyses predicting DDDI-RD and DDDI-AD in the reduced subsample (drivers holding a license for ≤ 10 years).

Regression model	Predictors	R^2	B	SE	Confidence interval 95%	
					Lower limit	Upper limit
Criterion: DDDI-RD		.12				
	Gender		-0.285*	0.034	-.352	-.218
	Age		-0.012*	0.004	-.019	-.006
	Experience		0.005*	0.001	.002	.007
	Gender x Age		0.003	0.004	-.006	.011
	Gender x Experience		-0.001	0.002	-.005	.002
	Age x Experience		0.000	0.000	-.001	.000
	Gender x Age x Experience		0.000*	0.000	.000	.001
Criterion: DDDI-AD		.05				
	Gender		-0.201*	0.041	-.282	-.118
	Age		-0.005	0.004	-.013	.003
	Experience		0.005*	0.001	.002	.007
	Gender x Age		0.000	0.005	-.010	.009
	Gender x Experience		-0.001	0.002	-.005	.002
	Age x Experience		0.000*	0.000	-.001	-.000
	Gender x Age x Experience		0.000	0.000	-.000	.001

* Confidence interval does not include zero.

Abbreviations: DDDI-RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI-AD = Dula Dangerous Driving Index Aggressive Driving subscale.

Finally, to ensure consistency with the analysis of the full sample, an additional analysis was conducted in the reduced subsample using years holding a driving license and annual car mileage as predictors. This analysis is provided in the supplementary material (see Section 2). Results were similar to those obtained in the full sample, with annual car mileage emerging as the relevant factor within driving experience.

3.3. DDDI-NCED subscale

In addition to the variables that constituted the focus of the present study, the DDDI includes the DDDI-NCED subscales that do not directly assess intentional dangerous driving

behaviors. For readers who may be interested, we briefly summarize the results for these variables below (for further details, see supplementary material, Section 1).

For the DDDI, both in the full sample and in the subsample of drivers holding a license for 10 years or less, the NCE subscale showed a significant Age $\times$ Annual car mileage/Experience $\times$ Gender interaction (95% *CI*s excluded zero). Among men, Age was negatively related to DDDI-NCED scores when Annual car mileage/Experience was at medium and high levels (95% *CI*s excluded zero). Conversely, Annual car mileage/Experience was positively related to DDDI-NCED scores when Age was low or medium (95% *CI*s excluded zero). Among women, the Age $\times$ Annual car mileage/Experience interaction was not significant (95% *CI*s included zero); however, Age and Annual Mileage/Experience were negatively and positively related, respectively, to DDDI-NCED scores (95% *CI*s excluded zero). Across all conditions, women reported significantly lower levels of DDDI-NCED scores than men (95% *CI*s excluded zero).

4. Discussion

The tendency to engage in dangerous driving behavior has been linked to a wide range of psychological and demographic factors, among which age and driving experience have been extensively studied as key predictors (Brown & Groeger, 1988; de Winter & Dodou, 2010; Huang & Winston, 2011; Jing et al., 2023; Jonah, 1986; Rowe et al., 2013). While increasing age is consistently associated with more cautious driving behavior, the impact of driving experience — and its relationship with age — remains inconclusive, with studies reporting mixed findings (Catchpole et al., 2005; de Winter & Dodou, 2010; Deery, 1999; Lee et al., 2008; McKenna et al., 2006; Waller et al., 2001). The present research aimed to clarify the relative contributions of these variables and to examine the potential interaction between age and driving experience in shaping the propensity to engage in unsafe driving behaviors. Additionally, the influence of gender was examined, given its well-established association with differences in driving style and risk-related tendencies (Cordellieri et al., 2016; Oltedal & Rundmo, 2006). To this end, two sets of analyses were conducted: one using the full sample, and another using a subsample of drivers holding a license for 10 years or less. The latter analysis was specifically designed to focus on the initial years of driving, a

period during which drivers typically have less experience and are still developing their driving skills.

The results of our study support the role of age as a predictor of dangerous driving behaviors. In line with our hypothesis (H1), age was negatively associated with DDDI-RD and DDDI-AD scores, suggesting that greater maturity contributes to a reduction in risky and aggressive driving behaviors (de Winter & Dodou, 2010; Hatfield & Fernandes, 2009; Jonah, 1990; Megías-Robles et al., 2022; Navon–Eyal & Taubman–Ben-Ari, 2020; Rhodes & Pivik, 2011). Importantly, this relationship remained significant after controlling for driving experience, a variable that is typically correlated with age. These findings are consistent with neurodevelopmental theories and psychological accounts proposing that, with increasing age, individuals exhibit greater self-regulatory capacity, including improved impulse control, enhanced emotional regulation, reduced sensation-seeking, and greater risk sensitivity (Hatfield et al., 2017; Huang & Winston, 2011; Navon-Eyal & Taubman-Ben-Ari, 2020; Pizzo et al., 2024; Rhodes et al., 2015; Steinberg, 2008).

Regarding driving experience, our findings indicated that annual car mileage was a more relevant indicator than the number of years holding a driving license. Annual car mileage was positively associated with the two assessed measures of dangerous driving behaviors, namely DDDI-RD and DDDI-AD. In contrast, the number of years holding a driving license was not significantly associated with any of these subscales. Within the subsample of drivers holding a license for 10 years or less, the composite Experience variable was also positively associated with DDDI-RD and DDDI-AD. These results are consistent with a strand of the existing literature suggesting that, as drivers gain experience, their increased perception of self-efficacy may foster riskier and aggressive driving attitudes and a higher frequency of traffic violations, driven by greater confidence in the skills acquired over time (Catchpole et al., 2005; de Winter and Dodou, 2010; Jing et al., 2023; Machado-León et al., 2016; Qu et al., 2014).

Taken together, these findings indicate that both age and driving experience (assessed by annual mileage) were significantly associated with dangerous driving behaviors, with each association remaining independent when controlling for the other. Accordingly, both factors should be considered when evaluating their respective contributions to driving behavior, a

topic that has been widely debated in the literature. However, the present study aimed to further our understanding of how age and experience relate to dangerous driving behavior by examining not only their individual associations but also their potential interaction, thereby providing a clearer insight into the nature of their interrelationship.

We hypothesized (H2) that driving experience could moderate age-related effects on driving behavior. The multiple regression analyses revealed a significant interaction between Age and Annual car mileage in predicting DDDI-RD and DDDI-AD, although for DDDI-RD, this interaction was observed only among men. In the subsample of drivers holding a license for 10 years or less, the interaction between Age and Experience was also significant for DDDI-RD (again, only among men) and DDDI-AD. Therefore, consistent with our second hypothesis, the results indicated that the previously discussed predictive roles of age and driving experience on dangerous driving were not present under all conditions. Specifically, higher age was associated with a lower tendency to engage in risky and aggressive behaviors only among drivers with sufficient experience. Among less experienced drivers, the frequency of such behaviors did not vary as a function of age. Furthermore, greater driving experience was associated with higher levels of dangerous behavior only among younger and middle-aged groups. In the oldest age group, DDDI scores were independent of experience. It is important to note that this interaction was observed only in men for the DDDI-RD subscale (see below for further details).

One possible explanation for the lack of a relationship between age and the tendency to engage in dangerous behaviors during the earliest stages of learning to drive is that, compared with more experienced drivers, novice drivers have not yet acquired many essential driving skills. These skills develop progressively along a learning curve guided by experience and include the automation of vehicle handling, visual scanning, attentional control, hazard detection, risk perception, and effective interaction with other road users, among others (Borowsky et al., 2010; Deery, 1999; Ge et al., 2017; McKnight & McKnight, 2003; Megias et al., 2018; Lehtonen et al., 2014). Drivers with limited skills may adopt a more cautious driving style due to low perceived self-efficacy. Consequently, younger drivers may not exhibit higher risk-taking than older drivers until they have developed sufficient driving proficiency. Once sufficient experience and perceived self-efficacy have been gained,

younger drivers would tend to demonstrate the pronounced risk-taking behavior characteristic of youth, particularly in men. From this perspective, the protective effect of age-related maturity on safer driving becomes evident only after a sufficient level of driving proficiency has been attained.

Our third hypothesis (H3) concerned gender differences in dangerous driving behavior. Men reported higher frequencies of DDDI-RD and DDDI-AD scores in the full sample, which is consistent with previous findings in the literature (Cordellieri et al., 2016; de Winter & Dodou, 2010; Megías-Robles et al., 2022; Oltedal & Rundmo, 2006). However, this gender difference was not observed in the reduced subsample of drivers holding a driving license for 10 years or less. An additional and noteworthy finding, highlighted earlier, is that the interaction between age and driving experience (or annual car mileage) was identified only among men for DDDI-RD. This pattern may be attributed to the lower overall risk-taking tendency observed in women compared to men.

Future research should aim to extend the present findings while addressing certain limitations of the current study. A primary concern is that the evaluation of driving behavior and experience relied on self-report measures, which are susceptible to social desirability bias and inaccuracies in individuals' perceptions of their own behavior. Future studies should therefore consider employing more naturalistic, objective, and fine-grained measures of driving behavior, such as driving simulators, telematics data, or official records of accidents and traffic violations. A further point to note is that, although assessing driving experience through both annual mileage and years licensed provides a more nuanced approach than simply categorizing drivers as inexperienced or experienced, it would be advisable to examine measures more specific to total lifetime mileage and compare them with the present data. Annual car mileage may fluctuate over time, particularly among older drivers, potentially affecting its validity as a measure of accumulated experience. A related consideration is that using mileage as an indicator of driving experience may introduce an exposure effect (Brown, 1982; Clarke et al., 2006), since greater distances travelled inherently increase the likelihood of engaging in dangerous behaviors. In the present study, however, the DDDI assesses behavior using relative frequency descriptors rather than absolute counts; therefore, exposure effects should not substantially influence the results.

Nevertheless, the use of more objective measures of risky driving behavior could help to further examine this issue.

Several additional points could also be considered in future research. First, a notable strength of the present study is that the sample was gender-balanced (52.57% men, 47.43% women) and encompassed a wide range of ages (18–70) and driving experience (up to 40 years); however, drivers with the greatest levels of experience were inevitably concentrated among the older participants. To address this limitation, we conducted an additional analysis focused exclusively on drivers in the early years of their driving trajectories and employed nonparametric methods to mitigate the effects of non-normal age distributions. Even so, future research would benefit from including a larger number of older drivers with relatively recent licenses, despite the practical challenges inherent in recruiting such a sample. Second, the use of longitudinal designs that track drivers over several years, particularly during the early stages of their driving careers, would provide valuable contributions to the literature. Finally, individual differences in psychological traits and cognitive abilities known to influence risky driving behavior, such as impulsivity, sensation seeking, emotion regulation, working memory, and visual attention, should be examined in conjunction with age and driving experience (Dahlen et al., 2005; Di Stasi et al., 2011; Megías et al., 2011, 2014; Ramos-Moreno et al., 2025; Underwood et al., 2003; Zhang et al., 2019; Zhang et al., 2023).

5. Conclusion

This study provides a more comprehensive understanding of how age and driving experience jointly relate to dangerous driving behavior, enabling clearer attribution of their respective influences. Drawing on a large, community-based, gender-balanced sample of drivers and using a recognized self-report measures of driving behavior (DDDI), our findings indicate that age and driving experience do not operate independently; rather, they interact in shaping dangerous driving tendencies. Age did not provide a uniform protective effect; its benefits became apparent only after drivers had acquired sufficient experience, as novice drivers may adopt a more cautious driving style due to their lack of driving skills and low perceived self-efficacy. Furthermore, higher levels of driving experience were associated with a greater frequency of dangerous behaviors, particularly among younger and intermediate age groups. These associations were not always consistent across genders, with

distinct patterns observed for men and women. Overall, the findings of this research underscore the importance of adopting multifactorial approaches to road safety that integrate demographic, experiential, and contextual factors. They can have direct implications for driver education, licensing policies, and prevention strategies, highlighting the need to develop more refined road safety interventions that take into account the combined roles of age, driving experience, and gender.

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Supplementary material

Section 1

Table S1.1. Means and standard deviations of the study variables for the DDDI-NCED questionnaire in the full sample and the reduced subsample (drivers holding a license for ≤ 10 years), for all participants and by gender.

	Mean (<i>SD</i>) Total sample	Mean (<i>SD</i>) Men	Mean (<i>SD</i>) Women
DDDI-NCED full sample	1.29 (0.56)	1.34 (0.59)	1.24 (0.53)
DDDI-NCED reduced subsample	1.33 (0.60)	1.39 (0.61)	1.28 (0.58)

Abbreviations: DDDI-NCED = Dula Dangerous Driving Index Negative Cognitive/Emotional Driving

Table S1.2. Bootstrapped Pearson correlation coefficients among the study variables for the DDDI-NCED subscale in the full sample, with *r* values and corresponding 95% confidence intervals reported (in square brackets).

	1	2	3
1. Age	—		
2. Years driving license	.90* [.88, .91]	—	
3. Annual car mileage	.13* [.09, .18]	.18* [.14, .22]	—
4. DDDI-NCED	-.09* [-.14, -.05]	-.05* [-.10, -.01]	.08* [.04, .13]

* Confidence interval does not include zero.

Abbreviations: DDDI-NCED = Dula Dangerous Driving Index Negative Cognitive/Emotional Driving

Table S1.3. Results for the multiple linear regression analyses predicting DDDI-NCED scores in the full sample.

Regression model	Predictors	R^2	b	SE	Confidence interval 95%	
					Lower limit	Upper limit
Criterion: DDDI-NCED		.04				
	Gender		-0.131*	0.041	-.211	-.054
	Age		-0.009*	0.003	-.015	-.002
	Years driving license		0.007	0.004	-.000	.015
	Annual car mileage		0.014*	0.007	.001	.027
	Gender x Age		0.002	0.004	-.006	.011
	Gender x Years driving license		-0.005	0.005	-.015	.004
	Gender x Annual car mileage		0.003	0.010	-.017	.022
	Age x Years driving license		-0.000	0.000	-.000	.000
	Age x Annual car mileage		-0.002*	0.000	-.003	-.001
	Gender x Age x Years driving license		0.000	0.000	-.000	.001
	Gender x Age x Annual car mileage		0.002*	0.001	.000	.003

* Confidence interval does not include zero.

Abbreviations: DDDI-NCED = Dula Dangerous Driving Index Negative Cognitive/Emotional Driving

Table S1.4. Bootstrapped Pearson correlation coefficients among the study variables for the DDDI-NCED subscale in the reduced subsample (drivers holding a license for ≤ 10 years), with r values and corresponding 95% confidence intervals reported (in square brackets).

	1	2	3	4	5
1. Age	—				
2. Years driving license	.37* [.31, .44]	—			
3. Annual car mileage	.05 [-.02, .11]	.19* [.12, .25]	—		
4. Experience	.27* [.21, .34]	.79* [.76, .81]	.63* [.58, .67]	—	
5. DDDI-NCED	-.12* [-.17, -.06]	.03 [-.04, .09]	.17* [.10, .23]	.08* [.02, .15]	—

* Confidence interval does not include zero.

Abbreviations: DDDI-NCED = Dula Dangerous Driving Index Negative Cognitive/Emotional Driving

Table S1.5. Results for the multiple linear regression analyses predicting DDDI-NCED scores in the reduced subsample (drivers holding a license for ≤ 10 years).

Regression model	Predictors	R^2	b	SE	Confidence interval 95%	
					Lower limit	Upper limit
Criterion: DDDI-NCED		.05				
	Gender		-0.099*	0.042	-.179	-.017
	Age		-0.012*	0.004	-.020	-.005
	Experience		0.006*	0.001	.004	.009
	Gender x Age		0.002	0.005	-.009	.013
	Gender x Experience		-0.006*	0.002	-.010	-.002
	Age x Experience		-0.001*	0.000	-.001	-.000
	Gender x Age x Experience		0.001	0.000	-.000	.001

* Confidence interval does not include zero.

Abbreviations: DDDI-NCED = Dula Dangerous Driving Index Negative Cognitive/Emotional Driving

Section 2

Table S2.1. Results for the multiple linear regression analyses predicting DDDI-RD in the reduced subsample (drivers holding a license for ≤ 10 years) including Years driving license and Annual car mileage as predictors.

Regression model	Predictors	R^2	b	SE	Confidence interval 95%	
					Lower limit	Upper limit
Criterion: DDDI-RD		.15				
	Gender		-0.262*	0.035	-.329	-.195
	Age		-0.010*	0.004	-.017	-.002
	Years driving license		0.013	0.010	-.005	.031
	Annual car mileage		0.042*	0.010	.023	.061
	Gender x Age		0.001	0.004	-.007	.009
	Gender x Years driving license		-0.012	0.012	-.035	.012
	Gender x Annual car mileage		0.007	0.013	-.032	.017
	Age x Years driving license		0.000	0.001	-.003	.003
	Age x Annual car mileage		-0.005*	0.001	-.008	-.002
	Gender x Age x Years driving license		0.000	0.001	-.003	.003
	Gender x Age x Annual car mileage		0.004*	0.002	.001	.007
Criterion: DDDI-AD		.08				
	Gender		-0.178*	0.041	-.258	-.097
	Age		-0.002	0.004	-.011	.007
	Years driving license		0.014	0.011	-.007	.035
	Annual car mileage		0.042*	0.011	.021	.063
	Gender x Age		-0.002	0.005	-.013	.007
	Gender x Years driving license		-0.004	0.014	-.031	.022
	Gender x Annual car mileage		-0.016	0.015	-.045	.013
	Age x Years driving license		-0.001	0.001	-.003	.003
	Age x Annual car mileage		-0.004*	0.002	-.008	-.001
	Gender x Age x Years driving license		0.000	0.002	-.003	.003
	Gender x Age x Annual car mileage		0.004	0.002	-.000	.007

* Confidence interval does not include zero.

Abbreviations: DDDI RD = Dula Dangerous Driving Index Risky Driving subscale; DDDI AD = Dula Dangerous Driving Index Aggressive Driving subscale.